

CHAPTER 1: ELECTRIC CHARGES AND FIELDS

Complete Class XII Physics Lecture Notes with Diagrams

1. Electric Charge

Electric Charge is an intrinsic property of matter that causes it to experience a force when placed in an electromagnetic field. It arises due to either the loss or gain of electrons by an atom.

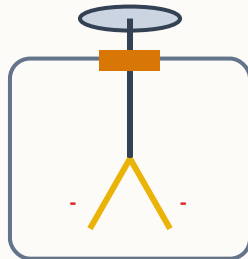
- **SI Unit:** Coulomb (C)
- **Elementary Charges:**
 - Charge on 1 electron (e^-) = $-1.6 \times 10^{-19} \text{ C}$
 - Charge on 1 proton (p^+) = $+1.6 \times 10^{-19} \text{ C}$
- There are only **two kinds** of electric charges: *Positive* and *Negative*.
- **Fundamental Law of Electrostatics:** Like charges repel each other, and unlike charges attract each other.

2. Electrostatic Induction

Electrostatic Induction is the process in which redistribution of electric charge takes place within a body when a charged body is brought near it (without physical contact).

Gold-Leaf Electroscope

A **Gold-Leaf Electroscope** is a sensitive instrument used to detect the presence and nature of electric charge on a body, operating on the principle of electrostatic induction and electrostatic repulsion between like charges.



Gold-Leaf Electroscope showing leaf divergence due to charge repulsion

3. Basic Properties of Electric Charge

i) Additivity of Electric Charge

Total charge of a system is the algebraic sum of all individual charges located at different points inside the system:

$$Q_{total} = q_1 + q_2 + q_3 + \dots + q_n$$

ii) Conservation of Charge

- The total electric charge of an isolated system always remains constant.
- Electric charges can neither be created nor destroyed; they can only be transferred from one body to another.

iii) Quantisation of Electric Charge

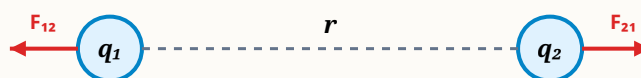
Electric charge cannot vary continuously to have any arbitrary value; it changes discontinuously to take any value from a discrete set. The experimental fact that electric charges occur in discrete amounts instead of continuous amounts is called **quantisation of electric charge**.

$$Q = \pm n e$$

where $n = 1, 2, 3, \dots$ (an integer) and $e = 1.6 \times 10^{-19} \text{ C}$

4. Coulomb's Law of Electrostatics

Coulomb's Law states that the magnitude of the electrostatic force between two stationary point charges is directly proportional to the product of the magnitudes of charges and inversely proportional to the square of the distance between them.



Electrostatic force between two point charges q_1 and q_2 separated by distance r

- $F \propto q_1 \times q_2$
- $F \propto 1 / r^2$

$$F = k \cdot (q_1 q_2 / r^2)$$

Where k is the electrostatic force constant, which depends on the medium.

- For free space (vacuum): $k = 1 / (4\pi\epsilon_0) = 9 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2$
- Here, ϵ_0 is the **permittivity of free space**:
 $\epsilon_0 = 8.854 \times 10^{-12} \text{ N}^{-1} \text{ m}^{-2} \text{ C}^2$ (or F/m)

$$F = (1 / 4\pi\epsilon_0) \cdot (q_1 q_2 / r^2)$$

Relative Permittivity (Dielectric Constant, K or ϵ_r):

$$K = \epsilon_r = \epsilon / \epsilon_0 = (\text{Permittivity of medium}) / (\text{Permittivity of free space})$$

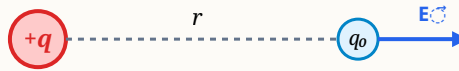
5. Electric Field (E)

Electric Field is the region around a charged body within which its electrostatic force can be experienced by other charges. Mathematically, it is defined as the electrostatic force experienced per unit positive test charge:

$$\vec{E} = \vec{F} / q_0$$

Electric Field due to a Point Charge

Consider a source charge q and a test charge q_0 separated by distance r :



Electric Field vector E produced by source charge $+q$ at distance r

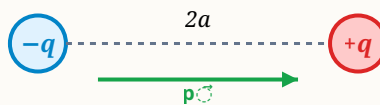
By Coulomb's Law: $F = k \cdot (q q_0 / r^2)$

Since $E = F / q_0$:

$$E = k \cdot (q / r^2) = (1 / 4\pi\epsilon_0) \cdot (q / r^2)$$

6. Electric Dipole

An **Electric Dipole** is a system of two equal and opposite point charges separated by a small distance ($2a$).



Electric Dipole with dipole moment vector $\vec{p}$ directed from $-q$ to $+q$

Electric Dipole Moment ($\vec{p}$)

It is the product of the magnitude of either charge and the distance between them:

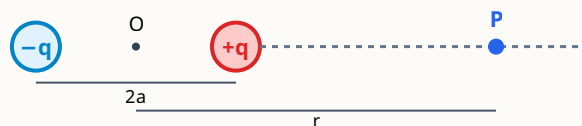
$$\vec{p} = q \times (2\vec{a})$$

- **Vector Quantity:** Direction is from *negative charge* ($-q$) to *positive charge* ($+q$).
- **SI Unit:** Coulomb-meter ($C \cdot m$).

7. Field of an Electric Dipole

A) Electric Field at an Axial Point of a Dipole

For a point P at distance r from the center of dipole along its axis:



Axial point P at distance r from dipole center O

- Electric field due to $+q$: $E_+ = k q / (r - a)^2$
- Electric field due to $-q$: $E_- = - k q / (r + a)^2$
- Net field $E_{net} = E_+ + E_- = k q [1/(r - a)^2 - 1/(r + a)^2]$

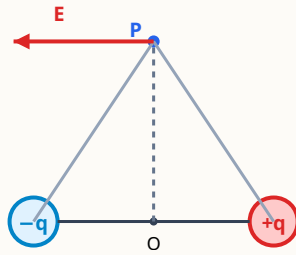
$$E_{axial} = 2 k p r / (r^2 - a^2)^2$$

For a short dipole ($a \ll r$, neglecting a^2):

$$E_{axial} = 2 k p / r^3 = (1 / 4\pi\epsilon_0) \cdot (2p / r^3)$$

B) Electric Field at an Equatorial Point of a Dipole

For a point on the perpendicular bisector at distance r from dipole center:



eq
Equatorial point P resolution of electric field components

$$E_{eq} = E_+ \cos\theta + E_- \cos\theta = 2 E_+ \cos\theta$$

$$E_{eq} = k p / (r^2 + a^2)^{3/2}$$

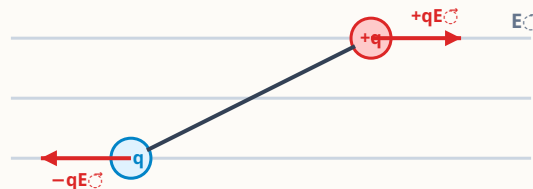
For a short dipole ($a \ll r$):

$$E_{eq} = k p / r^3 = (1 / 4\pi\epsilon_0) \cdot (p / r^3)$$

Key Relation: For a short dipole at equal distances, $E_{axial} = 2 \times E_{eq}$.

8. Torque on a Dipole in a Uniform Electric Field

When a dipole is placed in a uniform electric field $\vec{E}$ at an angle θ :



Couple of forces ($+qE$ and $-qE$) creating torque on a dipole in uniform field

Net force $F_{net} = qE - qE = 0$.

Torque $\tau = \text{Force} \times \text{Perpendicular distance} = qE \times (2a \sin\theta)$:

$$\tau = p E \sin\theta \implies \vec{\tau} = \vec{p} \times \vec{E}$$

Special Cases:

- $\theta = 0^\circ$: $\tau = 0 \rightarrow$ **Stable Equilibrium**
- $\theta = 90^\circ$: $\tau = pE$ (Maximum)
- $\theta = 180^\circ$: $\tau = 0 \rightarrow$ **Unstable Equilibrium**

9. Electric Field Lines

An **electric field line** is an imaginary continuous path/curve along which a small positive test charge would tend to move if free to do so. The tangent to the field line at any point gives the direction of the electric field at that point.



Electric Field Lines for isolated Positive (+q) and Negative (-q) charges

Properties of Electric Field Lines:

1. Field lines are continuous smooth curves without any breaks.
2. They start from positive charges and end at negative charges. They do not form closed loops.
3. The tangent at any point gives the direction of the electric field at that point.
4. **No two field lines can intersect each other.** Reason: If they intersect, there would be two tangents at the point of intersection, implying two different directions of electric field at a single point, which is impossible.
5. Field lines are always normal to the surface of a conductor in equilibrium.
6. **Relative Closeness:** Represents strength of electric field.
 - Close together $\Rightarrow$ Strong field
 - Far apart $\Rightarrow$ Weak field
 - Parallel & equally spaced $\Rightarrow$ Uniform electric field
7. **Electrostatic Shielding:** Field lines do not pass through a conductor because $E = 0$ inside a charged conductor.

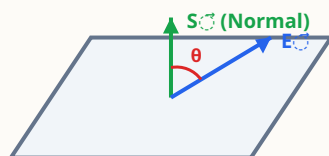
10. Area Vector & Electric Flux

Area Vector ($\vec{S}$)

Area of a surface is treated as a vector quantity. Its direction is defined along the **outward normal** to the surface.

Electric Flux (Φ_E)

Electric flux measures the total number of electric field lines passing normally through a given area:



Electric Flux through area element with angle θ between E and Normal S

$$\Phi_E = \vec{E} \cdot \vec{S} = E S \cos\theta$$

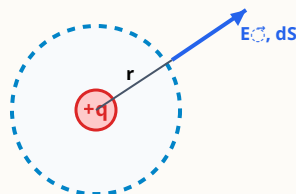
For infinitesimal area element $d\vec{S}$:

$$d\Phi_E = \vec{E} \cdot d\vec{S}$$

- SI Unit: $N \cdot m^2/C$ or $V \cdot m$
- Scalar Quantity

11. Gauss's Law / Theorem

Gauss's Theorem states that the total electric flux (Φ_E) through any closed surface (Gaussian surface) is equal to $1 / \epsilon_0$ times the total net charge enclosed by the closed surface.



Spherical Gaussian surface around point charge $+q$

$$\Phi_E = \oint \vec{E} \cdot d\vec{S} = q_{\text{enclosed}} / \epsilon_0$$

Gaussian Surface

Any hypothetical closed surface enclosing a charge is called the **Gaussian surface** of that charge.

Proof of Gauss's Theorem (for a Point Charge)

Consider a point charge $+q$ at the center of a spherical Gaussian surface of radius r :

1. Electric field at any point on spherical surface: $E = (1 / 4\pi\epsilon_0) \cdot (q / r^2)$
2. Angle between field vector $\vec{E}$ and surface element vector $d\vec{S}$ is $\theta = 0^\circ$.
3. Total Flux:

$$\Phi_E = \oint \vec{E} \cdot d\vec{S} = \oint E dS \cos(0^\circ) = E \oint dS$$

4. Since surface area of sphere $\oint dS = 4\pi r^2$:

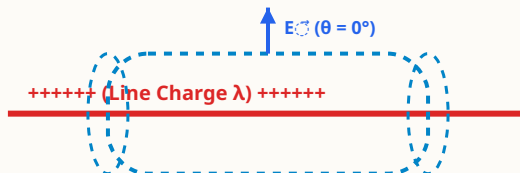
$$\Phi_E = [(1 / 4\pi\epsilon_0) \cdot (q / r^2)] \times 4\pi r^2 = q / \epsilon_0$$

$$\Phi_E = q / \epsilon_0 \quad (\text{Hence Proved})$$

12. Applications of Gauss's Law

A) Electric Field due to a Long Infinite Straight Wire

Consider a long thin wire with linear charge density $\lambda = q / l$. Construct a cylindrical Gaussian surface of radius r and length l around the wire:



Cylindrical Gaussian surface surrounding infinitely long charged wire

The cylindrical Gaussian surface has 3 parts:

1. Curved surface (S_2) where $\theta = 0^\circ$
2. Two flat circular end caps (S_1 and S_3) where $\theta = 90^\circ$

Applying Gauss's Law:

$$\oint \vec{E} \cdot d\vec{S} = q / \epsilon_0$$

$$\int E dS_1 \cos 90^\circ + \int E dS_2 \cos 0^\circ + \int E dS_3 \cos 90^\circ = q / \epsilon_0$$

$$0 + E \int dS_2 + 0 = q / \epsilon_0$$

Since curved surface area of cylinder $\int dS_2 = 2\pi r l$:

$$E \times 2\pi r l = q / \epsilon_0 \implies E = q / (2\pi r l \epsilon_0)$$

Substituting linear charge density $\lambda = q / l$:

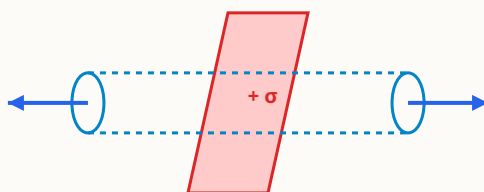
$$E = \lambda / (2\pi \epsilon_0 r)$$

Important Conceptual Note:

If a point charge q is placed on a vertex of a cube, then by symmetry, assuming this vertex as the center, it will require **8 identical cubes** to completely enclose the charge. Thus, flux through one cube is $\Phi = q / (8\epsilon_0)$.

B) Electric Field due to Uniformly Charged Infinite Plane Sheet

Consider an infinite plane sheet with surface charge density $\sigma = q / A$. Construct a pillbox cylindrical Gaussian surface passing through the sheet with cross-sectional area $A = \pi r^2$:



Pillbox Gaussian surface through an infinite thin charged sheet

The Gaussian surface consists of two flat end caps (S_1, S_3 where $\theta = 0^\circ$) and a curved side surface (S_2 where $\theta = 90^\circ$).

Applying Gauss's Law:

$$\int E dS_1 \cos 0^\circ + \int E dS_2 \cos 90^\circ + \int E dS_3 \cos 0^\circ = q / \epsilon_0$$

$$E \int dS_1 + 0 + E \int dS_3 = q / \epsilon_0$$

$$E (A) + E (A) = q / \epsilon_0 \implies 2 E A = q / \epsilon_0$$

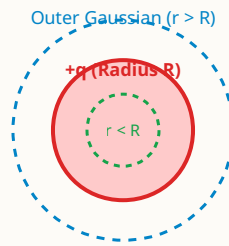
Since $q = \sigma A$:

$$2 E A = (\sigma A) / \epsilon_0$$

$$E = \sigma / (2\epsilon_0) \quad (\text{Constant, independent of distance } r)$$

C) Electric Field due to Uniformly Thin Spherical Shell

Consider a thin spherical shell of radius R carrying total charge q :



Concentric Gaussian surfaces inside ($r < R$) and outside ($r > R$) thin spherical shell

i) Inside the Shell ($r < R$):

Construct a spherical Gaussian surface inside the shell ($r < R$). Since charge inside a hollow conductor is zero ($q_{\text{enclosed}} = 0$):

$$\oint \vec{E} \cdot d\vec{S} = q / \epsilon_0 = 0 \implies E_{\text{inside}} = 0$$

ii) Outside the Shell ($r > R$):

Construct a spherical Gaussian surface of radius $r > R$ surrounding the shell:

$$\oint \vec{E} \cdot d\vec{S} = q / \epsilon_0$$

$$E \oint dS = q / \epsilon_0 \implies E \times (4\pi r^2) = q / \epsilon_0$$

$$E_{\text{outside}} = q / (4\pi \epsilon_0 r^2)$$

iii) On the Surface ($r = R$):

$$E_{\text{surface}} = q / (4\pi \epsilon_0 R^2)$$

13. Deduction of Coulomb's Law from Gauss's Law

Consider a point charge $+q$. Construct a spherical Gaussian surface of radius R around it. Place a test charge q_0 at point P on this surface.

According to Gauss's Law:

$$\Phi = q / \epsilon_0$$

$$\oint \vec{E} \cdot d\vec{S} = q / \epsilon_0$$

$$\oint E dS \cos 0^\circ = q / \epsilon_0$$

$$E \times 4\pi R^2 = q / \epsilon_0 \implies E = q / (4\pi \epsilon_0 R^2)$$

Force on test charge q_0 :

$$F = q_0 E$$

$$F = q_0 \times [q / (4\pi \epsilon_0 R^2)]$$

$$F = (1 / 4\pi \epsilon_0) \cdot (q q_0 / R^2) = K q q_0 / R^2$$

(This is Coulomb's Law, derived from Gauss's Law)